

Dual cotype forces the disjoint property

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2026-06-20

Abstract

Botelho, Garcia, and Miranda ask in Remark 2.8 of [1] whether there can be a Banach lattice E such that E^* has cotype p , $2 \leq p < \infty$, while E fails the disjoint property of order p . The answer is negative. If E^* has cotype p , then every bounded disjoint sequence in E is weakly p -summable. Hence no such counterexample exists.

1 Source question

The source is G. Botelho, L. A. Garcia, and V. C. C. Miranda, *Disjoint p -convergent operators and their adjoints*, arXiv:2309.03775.

The relevant passage is Remark 2.8 on page 5 of the source PDF:

for every $x \in X$. In particular, $q_T(\|x\| - \varepsilon) > \varepsilon$ for every $x \in X$, which proves that T is q_T -almost order bounded. Thus T is an almost Dunford-Pettis operator. $\square$

Remark 2.8. Let us discuss briefly the assumptions on E in the theorem above. On the one hand, we can apply the theorem for $E = c_0$ only for $p \geq 2$ ($c_0^* = \ell_1$ has cotype 2); but using that c_0 has the disjoint property of order 1 (Example 2.6(5)), we can apply the theorem for every $p \geq 1$. On the other hand, we still do not know whether there exists a Banach lattice E such that E^* has cotype $2 \leq p < \infty$ and E fails the disjoint property of order p .

In paraphrase, the authors ask whether dual cotype p can hold while the underlying Banach lattice fails the disjoint property of order p . Their Definition 2.5 says that E has the disjoint property of order p when every bounded disjoint E -valued sequence is weakly p -summable.

2 Proof intuition

The only missing bridge is order continuity. A Banach lattice with finite cotype cannot contain an order-bounded disjoint sequence whose norms stay away from zero: all signed finite sums are dominated in modulus by the same positive order bound, but the cotype inequality would force the ℓ_p sum of the norms to grow like $N^{1/p}$. Applied to E^* , this gives order-continuity of the dual norm.

Once E^* has order-continuous norm, every bounded disjoint sequence in E is weakly null. Such a sequence, together with its weak limit 0, is relatively weakly compact. The source paper's Lemma 2.4 then applies because the sequence lies in the solid hull of this weakly compact set.

3 The result

Lemma 1 (Finite cotype forces order-continuity). *Let F be a Banach lattice. If F has cotype $p < \infty$, then the norm of F is order continuous.*

Proof. Let C be a cotype p constant for F . Suppose that the norm of F is not order continuous. By the standard order-continuity criterion for Banach lattices, there are $0 \leq y_n \leq y$ with the y_n 's pairwise disjoint and, after passing to a subsequence, $\|y_n\| \geq \delta > 0$ for all n .

For every finite set $\{1, \dots, N\}$ and every choice of signs $\varepsilon_n = \pm 1$, disjointness gives

$$\left| \sum_{n=1}^N \varepsilon_n y_n \right| = \sum_{n=1}^N y_n \leq y,$$

and therefore $\|\sum_{n=1}^N \varepsilon_n y_n\| \leq \|y\|$. Applying the cotype inequality to $y_1, \dots, y_N$ yields

$$\delta N^{1/p} \leq \left(\sum_{n=1}^N \|y_n\|^p \right)^{1/p} \leq C \left(\int_0^1 \left\| \sum_{n=1}^N r_n(t) y_n \right\|^2 dt \right)^{1/2} \leq C \|y\|.$$

Letting $N \rightarrow \infty$ gives a contradiction. □

Theorem 1. *Let E be a Banach lattice and let $2 \leq p < \infty$. If E^* has cotype p , then E has the disjoint property of order p .*

Proof. By Lemma 1, the Banach lattice E^* has order-continuous norm. A standard characterization of order-continuity of the dual norm says that every norm-bounded disjoint sequence in E is weakly null; this is the criterion cited as Theorem 2.4.14 of Meyer-Nieberg in the source paper.

Let (x_n) be any bounded disjoint sequence in E . The preceding paragraph gives $x_n \rightarrow 0$ weakly. Hence

$$W = \{0\} \cup \{x_n : n \in \mathbb{N}\}$$

is weakly compact. The sequence (x_n) is contained in the solid hull $\text{sol}(W)$, and it is disjoint. Lemma 2.4 of [1] states that, when E^* has cotype p , every disjoint sequence contained in the solid hull of a relatively weakly compact subset of E is weakly p -summable in E . Therefore (x_n) is weakly p -summable.

Since (x_n) was an arbitrary bounded disjoint sequence, E has the disjoint property of order p . □

Corollary 1 (Answer to Remark 2.8). *There is no Banach lattice E such that E^* has cotype $2 \leq p < \infty$ and E fails the disjoint property of order p .*

Proof. This is exactly Theorem 1. □

4 Verification and novelty check

The proof has no computational dependency. The crop was rendered from the arXiv PDF with `code/crop_open_problem.py`.

For duplicate and novelty checking, the run indexes were searched for 2309.03775, the source title, `disjoint property of order`, `dual cotype`, and `weakly p-summable`. No previous run packet for this question was found. Web searches on 2026-06-20 for the source question, for `E*` has

cotype together with disjoint property of order p , and for close phrases involving bounded disjoint weakly p -summable sequences did not locate a separate later resolution. The proof above is a short consequence of the source paper's Lemma 2.4 together with the standard finite-cotype/order-continuity observation proved in Lemma 1; human review should decide whether to describe the result as a new observation or as an implicit consequence of already available tools.

5 Human review recommendation

Recommended review focus:

- Check that the non-order-continuity criterion used in Lemma 1 is the intended Banach-lattice norm criterion.
- Check that Lemma 2.4 of [1] is applied with exactly the same cotype exponent p .
- Check that the weakly null sequence plus its limit is accepted as a weakly compact set in the weak topology.

If these standard points are accepted, the source question is fully settled with a negative answer.

References

- [1] G. Botelho, L. A. Garcia, and V. C. C. Miranda, *Disjoint p -convergent operators and their adjoints*, arXiv:2309.03775, 2023.
- [2] C. D. Aliprantis and O. Burkinshaw, *Positive Operators*, Springer, Dordrecht, 2006.
- [3] P. Meyer-Nieberg, *Banach Lattices*, Springer-Verlag, Berlin, 1991.